COMP 1800

Data Visualization



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Introduction to Data Visualisation

Data visualisation can be thought of as the process of telling a story with data. Rather than just working with raw numbers we use a range of techniques to present data visually in a way that highlights meaningful insights. These insights can then support decision making, helping organisations develop strategies, identify problems, and drive improvements or change. In simple terms, data visualisation is about turning complex data into something that is easier to understand and act upon.

Visualisation mainly serves two purposes exploration and explanation. Exploration involves using visual tools to understand the structure, patterns, and key characteristics of a dataset and explanation focuses on clearly presenting insights after analysis. Exploration forms the foundation for deeper understanding. This process often requires some level of data preparation including data cleaning and data wrangling, where data is transformed into a suitable format using techniques like binning, segmentation, and clustering. While information visualisation is closely related and refers to the visual representation of abstract data to support human understanding, it is a broader concept that includes areas like knowledge visualisation. In this module the focus remains specifically on data visualisation as a tool for analysing and communicating data effectively, where tools like Excel, Tableau, and Power BI are useful for quick visualisation, but Python or R are preferred as they allow reusable scripts that can be applied across multiple datasets.

Discussion

The analysis revealed that the venues can be clearly segmented into low, medium, and high-volume groups based on their total number of visitors. A small number of venues dominate in terms of visitor volume while most venues attract only moderate or low numbers of visitors. This long-tail pattern appeared consistently across several visualisations and suggests that visitor demand is unevenly distributed between venues.

The correlation analysis showed that visitor numbers are strongly related to the distance travelled by visitors, indicating that the busiest venues tend to attract people from further away. The relationship between visitor numbers and average spend was slightly negative. This was one of the most important findings in the analysis because it showed that the venues attracting the largest crowds are not necessarily those generating the highest spending per visitor. Instead the highest-spending venues tended to have lower visitor numbers, longer visit durations, older visitors and a higher proportion of female visitors.The radar chart and line plot both supported this finding. The venues with the highest average spend shared several characteristics, including relatively low visitor numbers, high female percentages, high average age and long visit durations. This suggests that average spend is influenced more by the type of visitors and how long they stay than by the total number of people visiting the venue. The analysis potrayed that there are different types of successful venues: some are successful because they attract many visitors, while others are successful because each visitor spends more.

The temporal analysis also revealed clear weekly and seasonal patterns. Sundays, Saturdays, and Mondays were the busiest days while Wednesdays and Thursdays had the lowest visitor numbers. Across the year, visitor numbers were generally highest in January, November, and December and lower during the summer months. This indicates that the venues experience both weekly and seasonal cycles in demand.Finally, the analysis identified several anomalies and operational changes. Some venues opened later during the year while others closed before the end of the period. These changes are important because they affect total visitor numbers and help explain why some venues appear to perform less strongly. A small number of unusual daily visitor values were also identified particularly in the busiest venues.

Visualisation 1

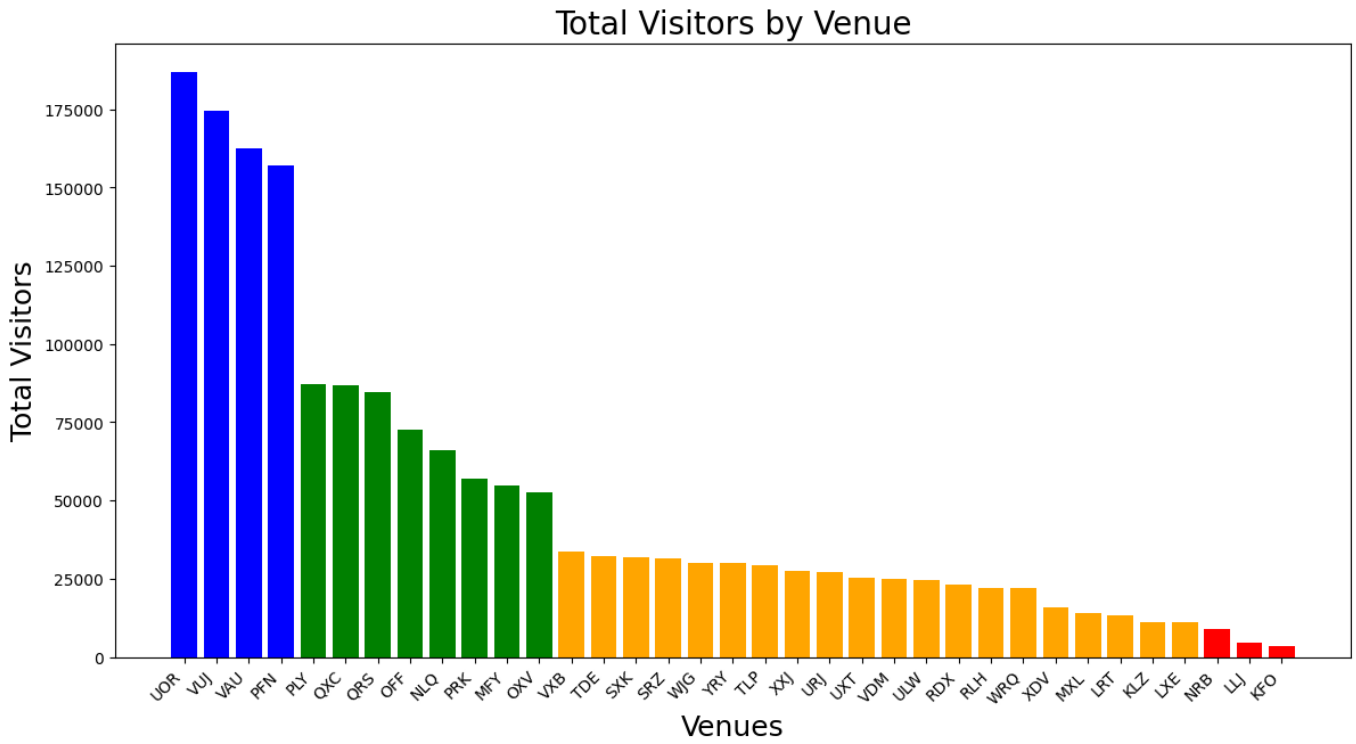


Figure 1. Total Visitors by Venue Bar chart

This visualisation was included to provide a clear comparison of total visitor volumes across all venues making it easy to identify which locations have the most visitors. The implementation follows the structure of the "L02 Proportion / 07BarChart all coloured.py and 10 BarChart grouped sorted.py." The original code was adapted and segmentation limits were defined based on the total number of visitors in each venue, where 0 to 10,000 represents Very Low, 10,000 to 40,000 Low, 40,000 to 100,000 Medium, and above 100,000 High. The colour coding separates these categories making it easier to clearly distinguish between high, medium, low, and very low volume venues.

The visualisation reveals a highly uneven distribution of visitors across venues. A small number of venues dominate with significantly higher visitor counts, indicating they are the most popular locations. In contrast, a large proportion of venues fall into the low and very low categories (orange and red), suggesting lower demand. The gradual decline in bar heights from left to right highlights a long-tail distribution, where only a few venues attract very high traffic while many receive moderate to minimal visitors. This visualisation is important because it raises further exploration questions, such as why those top venues attract so many visitors and what causes the significant drop across others. It also reflects how the data was segmented using thresholds based on total visitors for each venue, following the same approach demonstrated in the lecture examples.

Visualisation 2

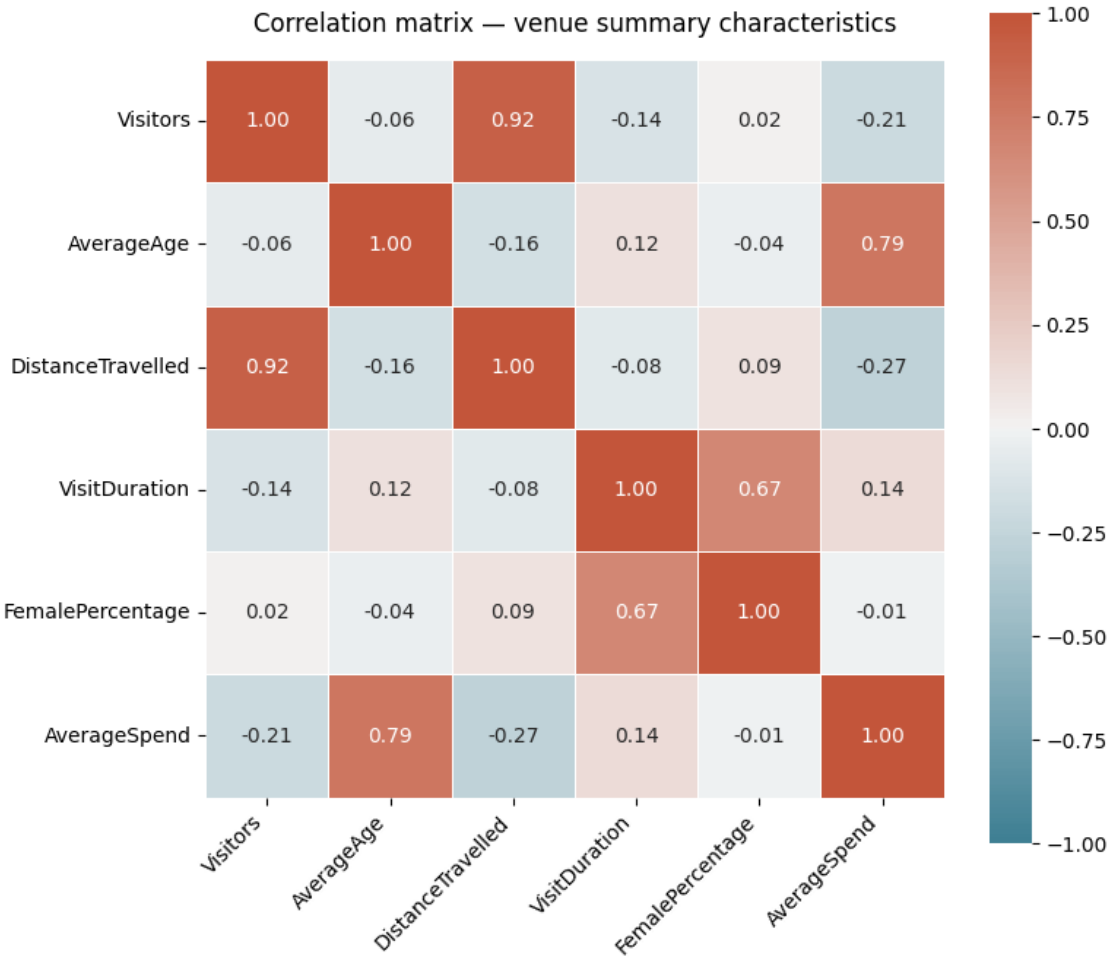


Figure 2. Heatmap Summary Correlation

This visualisation was included to examine the relationships between key venue characteristics and identify how different variables influence one another. Understanding these relationships is important for uncovering patterns that are not immediately visible from raw data, such as whether visitor numbers are linked to spending behaviour or travel distance. The implementation was based on the lecture example "L04 Correlation / 12Heatmap summary correlation.py". The code was adapted with venue-specific attributes such as total visitors, average age, distance travelled, visit duration, gender distribution, and average spend. Formatted annotations were added to improve readability and make the visualisation more suitable for analysis.

The heatmap reveals several important relationships between the variables. The strongest positive correlation is observed between visitor numbers and distance travelled (0.92), indicating that venues attracting more visitors tend to draw people from greater distances. There is also a strong positive relationship between average age and average spend (0.79), suggesting that older visitors generally spend more. Another notable pattern is the moderate positive correlation between visit duration and female percentage (0.67), which implies that venues with a higher proportion of female visitors tend

to have longer visit durations.In contrast, some variables show weak or negligible relationships. For example, female percentage has almost no correlation with average spend (-0.01), indicating that spending behaviour is largely independent of gender distribution. Visitor numbers show a slight negative correlation with average spend (-0.21), suggesting that busier venues may not necessarily generate higher spending per visitor. Overall, the visualisation highlights both strong and weak relationships, helping to identify the most influential factors affecting visitor behaviour across venues. This visualisation now leads us to explore these correlations and provide some new insights.

Visualisation 3

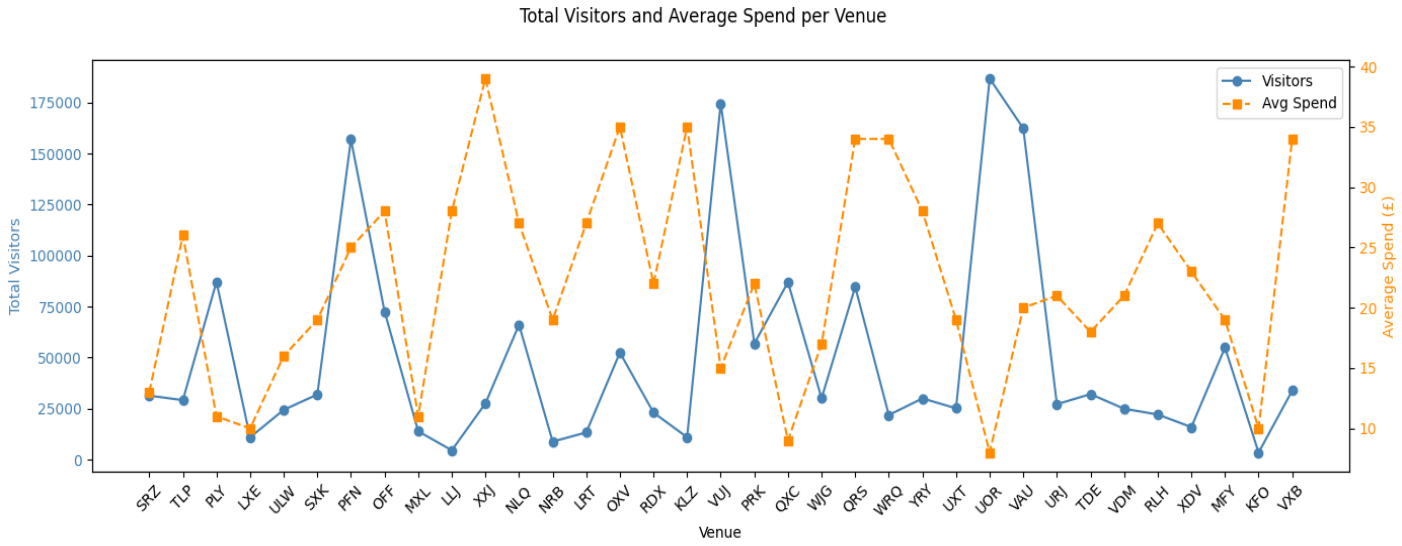


Figure 3.Line Plot of Total Visitors and Average Spend per Venue

This visualisation was included to compare how total visitor numbers and average spending vary across different venues, allowing patterns between volume and spending behaviour to be observed simultaneously. A line plot was chosen because it clearly shows fluctuations and relative differences across categories, making it easier to identify peaks and drops in both variables. The implementation was based on the lecture example “L03 Time / 02Line Plot high volume.py”, where multiple series are plotted on the same axes. The code was adapted by introducing a secondary y-axis using twinx() to accommodate the different scales of visitors and average spend. Additional styling, such as markers, colours, and rotated axis labels, was applied to improve readability and distinguish between the two variables.

This visualisation was motivated by observing the negative correlation between average spend and visitor numbers which highlights an important insight the venues with the highest number of visitors are not necessarily the ones generating the most revenue. The average spend appears to be higher in lower-volume venues suggesting that smaller or less crowded locations may attract visitors who are willing to spend more. Building on this, the visualisation reveals a clear variation in visitor numbers across venues, where only a few venues experience very high visitors while the majority remain at moderate or low levels clarifying the uneven distribution seen earlier. At the same time, average

spend fluctuates independently, and its pattern does not consistently match visitor trends. This means that even when a venue is highly popular, it does not guarantee higher spending per visitor. Overall the lack of alignment between the two lines supports the earlier correlation findings and suggests that visitor volume and spending behaviour are influenced by different factors, making this a key insight in understanding venue performance.

Visualisation 4

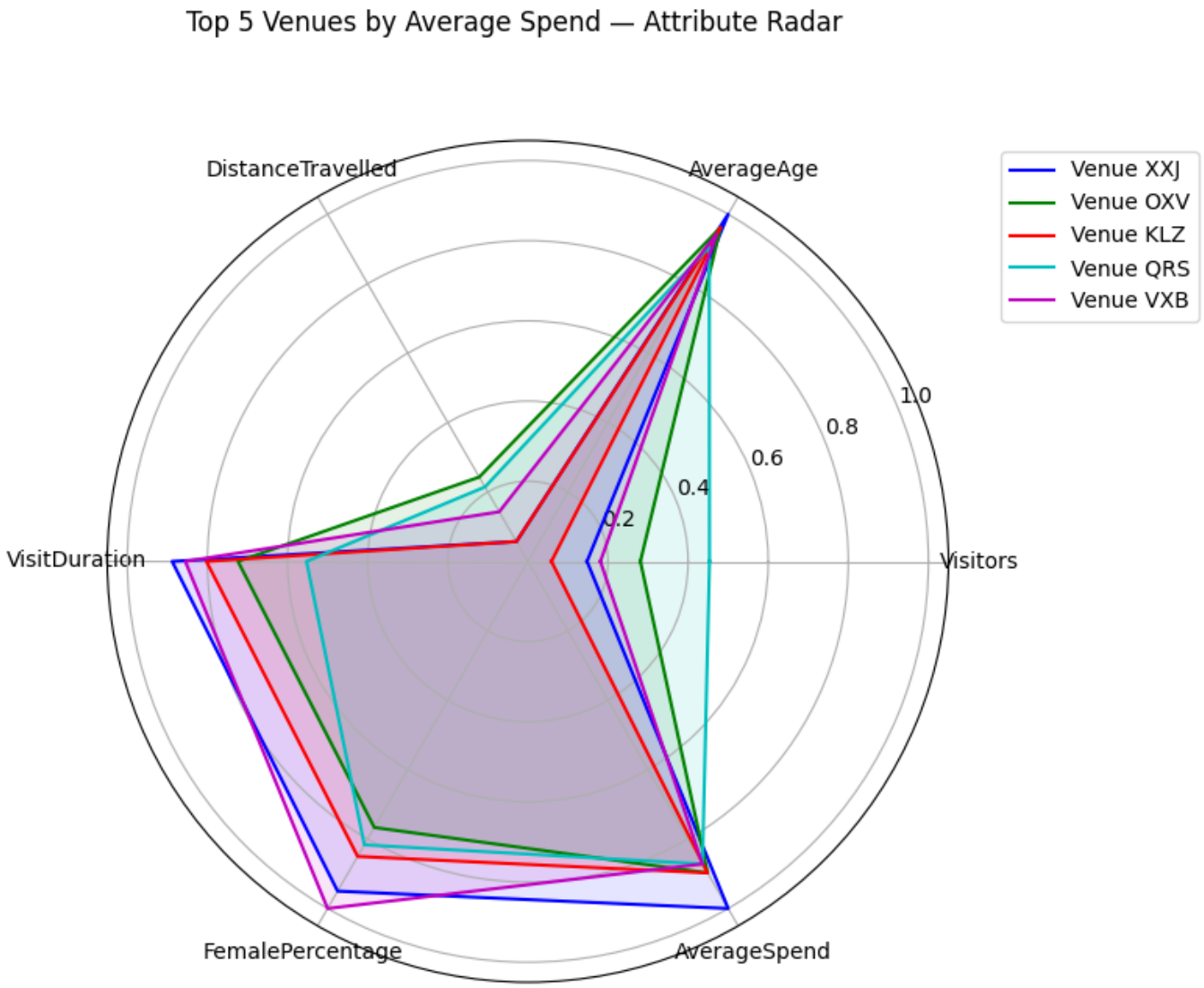


Figure 4. Radar plot Top Venues by Average spend

This visualisation was based on the need to look more closely at the venues with the highest average spend and understand whether they also share similar characteristics in other areas. While earlier visualisations showed spending patterns on their own, this radar chart makes it possible to compare several attributes for the same venues at once, including visitors, average age, distance travelled,

visit duration, female percentage, and average spend. The code was adapted from the lecture “L06 Dimension / 06RadarPlot high volume overlaid.py”. The data was normalised in the same way to allow fair comparison across different scales and the selected items were changed to the top five venues by average spend that was found out by sorting average spend in decending order.

This visualisation is important because it was used to identify what the venues with the highest average spend have in common. Rather than looking at spending alone the radar plot allows all of the main venue characteristics to be compared at the same time making it easier to see the shared patterns between the top five venues by average spend. The plot shows that these venues tend to have a high proportion of female visitors, longer visit durations and older average visitor ages. These patterns are particularly interesting because in figure-2 heatmap visit duration and female percentage showed a positive relationship, while average age also showed a strong positive relationship with average spend. This suggests that these characteristics may play an important role in explaining why these venues generate higher spending per visitor. The radar chart also reveals what these venues do not have in common none of the top-spending venues has a particularly high number of visitors or a high distance travelled value. This reinforces the earlier finding that the busiest venues are not necessarily the most profitable on a per-visitor basis. Instead, the venues with the highest average spend appear to attract smaller numbers of visitors who stay longer, are older on average, and are more likely to be female. Similarly, the relatively low distance travelled values suggest that venues attracting more local visitors may generate greater spending than those drawing people from further away.

Visualisation 5

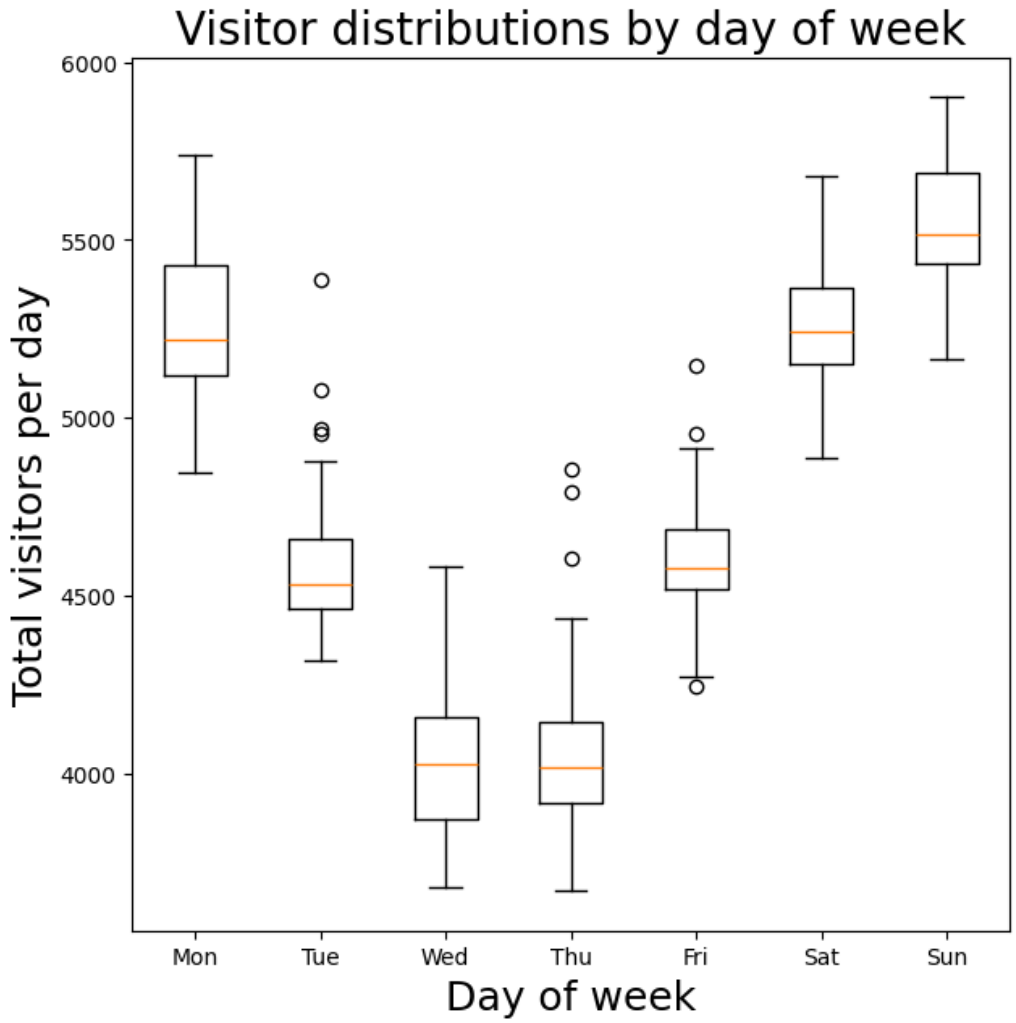


Figure 5. Visitor Distribution by Day of Week

This visualisation was included to investigate whether visitor numbers vary systematically depending on the day of the week. Since earlier visualisations focused mainly on differences between venues, it was also important to explore whether there are broader patterns affecting overall visitor behaviour. The implementation was adapted from the lecture example "L05 Distribution / 06Box Plot low volume.py". The code was modified by first summing the total number of visitors across all venues for each day and then grouping these totals according to the day of the week. A box plot was chosen because it not only shows the typical visitor level for each day, but also reveals the range, variability, and presence of unusually high or low values.

The box plot reveals a clear weekly pattern in visitor behaviour. Sunday has the highest visitor numbers with the median visitor count noticeably above every other day, followed by Saturday and Monday. In contrast, Wednesday and Thursday have the lowest typical visitor numbers, suggesting that midweek is the quietest period. Friday and Tuesday sit between these two extremes, although both show a wider spread and several unusually high values. The presence of outliers on days such as Tuesday, Thursday, and Friday indicates that there are occasional days when visitor numbers rise

far above the normal level. The visualisation shows that visitor demand changes considerably across the week, with weekends attracting the largest crowds and midweek attracting the fewest visitors.

Visualisation 6 (Interactive)

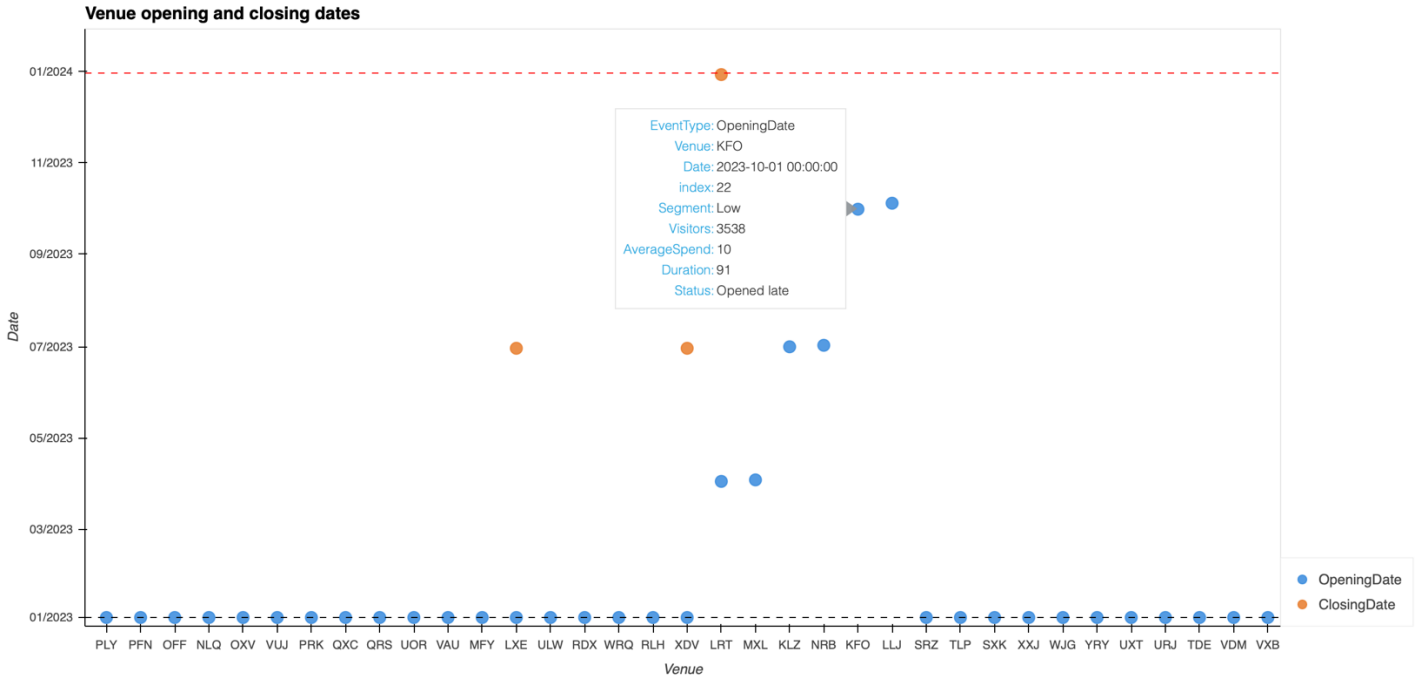


Figure 6. Bubble plot Venue Opening and Closing Dates

This visualisation was included to explore whether all venues operated throughout the full time period or whether some opened later or closed. This is important because differences in opening and closing dates can affect the total visitor numbers recorded for each venue meaning some venues may have lower totals simply because they were active for a shorter period. The visualisation was adapted from the lecture example “L07 Interaction /15BubblePlot marketing sales profit limits.py”. The code was changed substantially to detect the first and last non-zero visitor dates for each venue, classify venues by operating status, and then reshape the data into long form so that opening and closing dates could both be plotted. Dashed reference lines were added to show the start and end of the full dataset period more clearly.

The visualisation shows that most venues were open for the full period, as their opening dates begin at the start of the dataset and no early closing date is shown. However, a small number of venues opened later, including **MXL, LRT, NRB, KLZ, KFO, and LLJ**, meaning they were only active for part of the year. This is important because their lower total visitor counts may partly reflect reduced operating time. The chart also shows that some venues closed including **LXE, XDV, and LRT**, with LXE and XDV closing around the end of June and LRT closing at the end of December. LRT is particularly notable because it both opened late and closed before the full end of the period, giving it the shortest effective operating window among the venues shown. Overall, this visualisation

provides important context because it shows that differences in visitor totals are not explained only by popularity, but also by how long each venue was open during the year.

Visualisation 7 (Interactive)

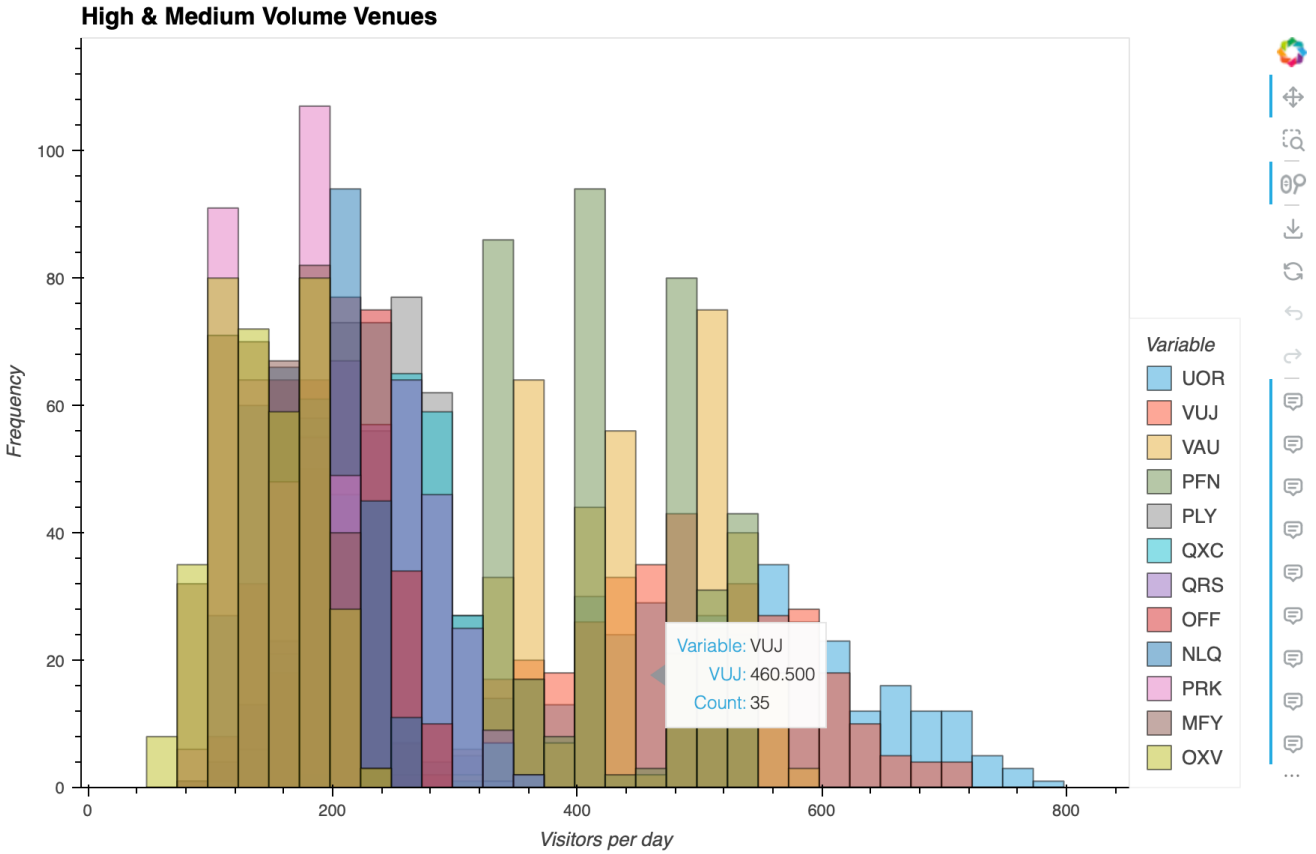


Figure 7.Daily Visitor Frequency Distribution — High & Medium Volume Venues

This visualisation was included to compare the daily visitor distributions of the highest and medium-volume venues rather than simply comparing their total yearly visitor counts. While earlier charts showed which venues attracted the most visitors overall, they did not reveal whether those venues were consistently busy every day or whether their totals were driven by occasional spikes. The implementation was adapted from the lecture example “L07 Interaction/ 07Histogram high volume custom bins.py”. The code was modified with the four highest-volume venues and a group of medium-volume venues. The histogram bins were also recalculated based on the range of visitor numbers in the venue dataset so that the different distributions could be compared more clearly.

The histogram shows that the high-volume venues generally have visitor distributions centred at much higher values than the medium-volume venues confirming that they attract larger crowds on most days rather than only occasionally. In particular, UOR, VUJ, VAU, and PFN all have distributions concentrated between roughly 350 and 600 visitors per day, with UOR extending furthest to the right and occasionally reaching more than 700 visitors. This suggests that UOR is not only one of the busiest venues overall, but also the most variable, experiencing both very high and more moderate

visitor days. In contrast, the medium-volume venues such as QXC, QRS, OFF, NLQ, PRK, MFY, and OXV are mostly concentrated between 100 and 300 visitors per day. Their distributions overlap considerably showing that these venues have relatively similar daily visitor patterns. The chart also reveals that some medium-volume venues occasionally reach visitor levels similar to the lower end of the high-volume venues, but they do so far less frequently. Overall the visualisation shows that the difference between high- and medium-volume venues is not just in their total number of visitors but also in the typical daily levels and variability of visitor demand

Visualisation 8

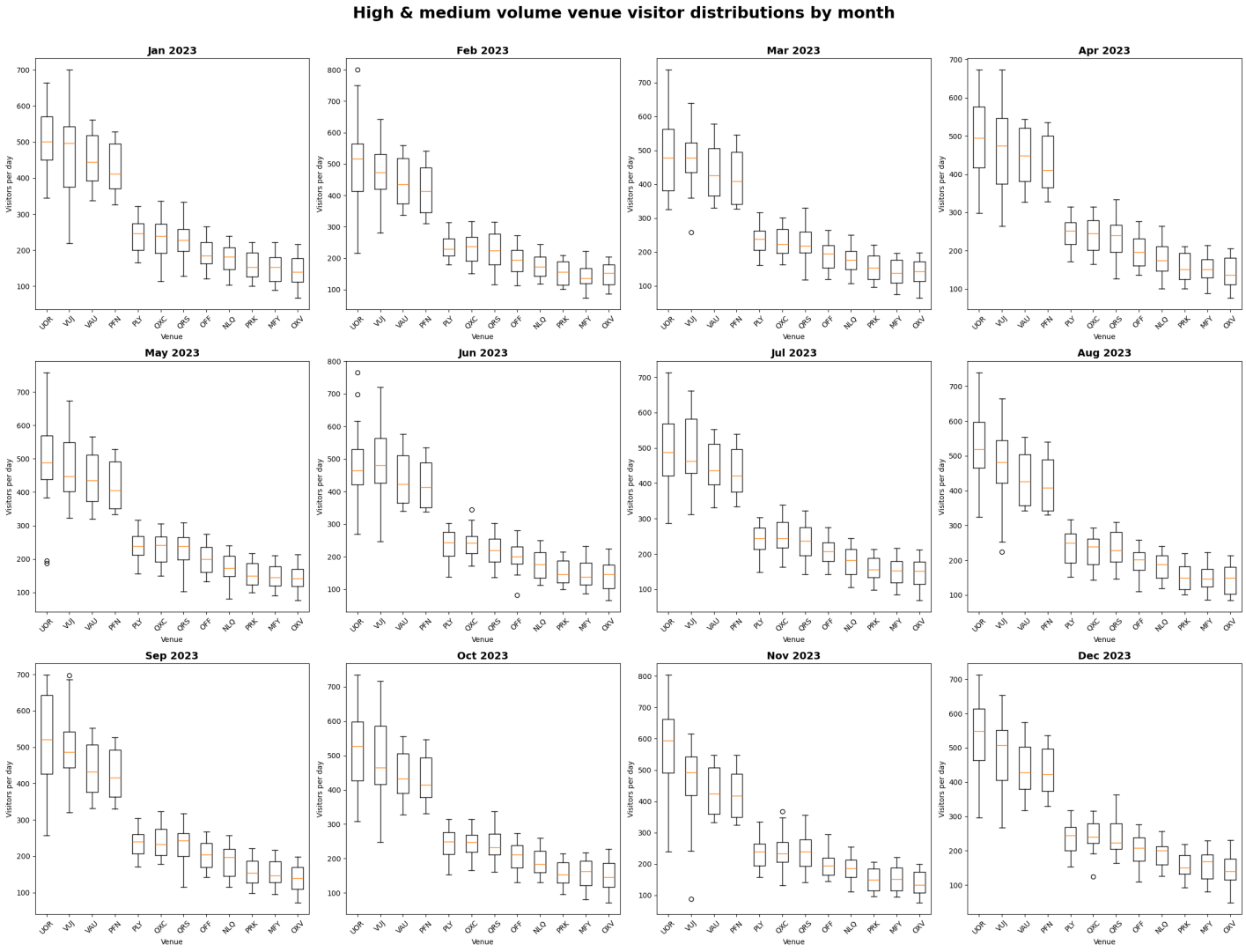


Figure 8Figure 4: Monthly Visitor Distribution by Venue — High & Medium Volume (Subplot).

This visualisation was included to investigate whether the daily visitor distributions of the high- and medium-volume venues change throughout the year, rather than remaining constant. Earlier visualisations showed the overall differences between these venues, but they did not reveal whether

certain months are consistently busier or quieter. The implementation combined ideas from the lecture examples “L05 Disteibution/ 04BoxPlot high volume.py” and “L04 Correlation / 02ScatterPlot medium volume subplots.py”. The box plot example was adapted while the subplot example was used to display a separate box plot for each month. The selected venues were the four highest-volume venues together with the main medium-volume venues, allowing changes in both groups to be compared side by side across the year.

The monthly box plots show that the overall ranking of the venues remains fairly consistent throughout the year, with UOR, VUJ, VAU, and PFN remaining the busiest venues in almost every month. However, there is also clear seasonal behaviour. Visitor numbers are generally higher at the start and end of the year, particularly in January, November, and December, where the medians and upper ranges for the busiest venues are noticeably larger. In contrast the summer months, especially June through August show lower visitor numbers and less variability for many venues. This suggests that demand falls during the middle of the year before increasing again toward the end. The medium-volume venues also follow this pattern, although their changes are smaller and they remain consistently below the high-volume venues. The box plots further reveal that some venues experience occasional unusually high or low days in particular months, shown by the outliers. For example, UOR has several unusually high visitor days in February and June, while some of the medium-volume venues show isolated low days. Overall, the visualisation indicates that the venue data is influenced not only by the type of venue but also by a clear seasonal cycle across the year.

Critical Review

This coursework demonstrated how the techniques taught in the module can be used to investigate a real dataset and communicate findings visually. Throughout the work, the main ideas from the module were applied, including data preparation, summary statistics, correlation analysis, segmentation, and the use of a range of visualisation techniques. The original CSV files were first combined into two structured dataframes one containing the daily visitor data and one containing summary information for each venue. This made it possible to explore the dataset from different perspectives and to reuse the same summary data in multiple visualisations.

A wide variety of visualisation types was used, including heatmaps, line plots, radar charts, box plots, histograms, scatter plots, and interactive plots. Each visualisation was selected because it was appropriate for answering a particular question about the data. For example, the heatmap was used to identify relationships between variables, the box plots were used to investigate variation and outliers, and the radar chart was used to compare the characteristics of the highest-spending venues. This reflects one of the key lessons from the module different visualisation methods are useful for different types of analysis.

The coursework also demonstrated best practices in visualisation design. Appropriate titles, axis labels, legends, and colour schemes were included so that each figure could be understood clearly without requiring additional explanation. Where appropriate, variables were normalised before

plotting so that different scales could be compared fairly. Interactive elements were used in some visualisations to make it easier to explore the data in more detail. The work also followed good analytical practice by not relying on a single figure alone. Instead, findings were checked across several visualisations, which increased confidence in the conclusions. For example, the negative relationship between visitor numbers and average spend first appeared in the heatmap and was then explored further using the line plot and radar chart.

The segmentation into low, medium, and high-volume groups was based only on visitor totals, which is simple and useful but we could have also used quantile based segemntation or other more efficient techniques.

Summary of Conclusions

The visualisations were designed to tell a clear analytical story, where each figure led naturally to further questions and deeper exploration. Findings from one visualisation were then investigated and supported through subsequent analyses, allowing the insights to be refined, validated, and used to guide the next stage of the investigation

- Visitor numbers are distributed unevenly, with only a few venues attracting very large numbers of visitors.
- Venues can be effectively segmented into low, medium, and high-volume groups based on total visitor numbers.
- The busiest venues are not the same as the highest-spending venues.
- Average spend tends to be higher in venues with:
    - Older visitors
    - Longer visit durations
    - Higher proportions of female visitors
    - Lower overall visitor numbers
- Visitor numbers and average spend show a slight negative relationship.
- The busiest venues tend to attract visitors from greater distances.
- Sundays and Saturdays are the busiest days, while Wednesdays and Thursdays are the quietest.
- Visitor numbers are generally highest in January, November, and December and lowest during the summer months.
- Several venues opened later or closed before the end of the period, which affects their total visitor numbers.

- Some venues show unusual spikes or drops in visitor numbers, indicating possible special events or other anomalies.

Declaration of AI Use

I have used AI while undertaking my assignment in the following ways:

- To explain concepts – YES
- To support my use of language – YES